LINUX ADMINISTRATION LAB SHEET — 2

Q1. Create a file named notes.txt and display its permissions string using the ls -l command.

```
shivangini@ubuntu:~$ touch notes.txt
shivangini@ubuntu:~$ ls -l notes.txt
```

Q2. Change the permissions of notes.txt to 644 using numeric (octal) mode, and verify the changes using ls -l.

```
shivangini@ubuntu:~$ chmod 644 notes.txt
shivangini@ubuntu:~$ ls -l notes.txt
```

Q3. Change notes.txt permissions to 600 (owner read/write only, no permissions for group and others) and verify.

```
shivangini@ubuntu:~$ chmod 600 notes.txt
shivangini@ubuntu:~$ ls -l notes.txt
```

Q4. Create an executable file named script.sh and grant execute permission exclusively to the owner using symbolic mode.

```
shivangini@ubuntu:~$ touch script.sh
shivangini@ubuntu:~$ chmod u+x script.sh
shivangini@ubuntu:~$ ls -l script.sh
```

Q5. Use the whoami command and ls -l notes.txt to identify the user and group ownership of notes.txt.

```
shivangini@ubuntu:~$ whoami
shivangini@ubuntu:~$ ls -l notes.txt
```

Q6. Run the ps command in your terminal and record how many processes are active in your current shell session.

```
shivangini@ubuntu:~$ ps
```

Q7. Run ps aux and count or estimate the total number of running processes across the entire operating system.

```
shivangini@ubuntu:~$ ps aux
shivangini@ubuntu:~$ ps aux | wc -l
```

Q8. Start a background process using sleep 60 & and immediately run jobs to verify that it is active in the background.

```
shivangini@ubuntu:~$ sleep 60 &
shivangini@ubuntu:~$ jobs
```

Q9. Use the kill command with the Process ID (PID) to terminate the active sleep process created in Exercise 8.

```
shivangini@ubuntu:~$ jobs -l
shivangini@ubuntu:~$ kill PID
```

Note: Replace PID with the actual process ID displayed by jobs -l.

Q10. Search for an available software package named tree using your system's package manager without installing it.

shivangini@ubuntu:~$ apt search tree

Q11. Display the current umask value of your terminal shell environment.

shivangini@ubuntu:~$ umask

Q12. Use symbolic chmod syntax to remove write permissions from the group on a file named document.txt.

shivangini@ubuntu:~$ touch document.txt
shivangini@ubuntu:~$ chmod g-w document.txt
shivangini@ubuntu:~$ ls -l document.txt

Q13. Bring a background job with Job ID 1 back to the foreground using the fg command.

shivangini@ubuntu:~$ fg %1

Q14. Send a background job with Job ID 1 to resume execution in the background using the bg command.

shivangini@ubuntu:~$ bg %1

Q15. Display the kernel information and system architecture using uname -a.

shivangini@ubuntu:~$ uname -a

Q16. Display a snapshot of all processes running for your user account using ps -u $USER.

shivangini@ubuntu:~$ ps -u $USER

Q17. Use pkill to terminate a test process running under the name sleep.

shivangini@ubuntu:~$ sleep 300 &
shivangini@ubuntu:~$ pkill -x sleep

Note: Run this only when you intend to terminate your test sleep process.

Q18. Check the current repository package index updates on an Ubuntu/Debian system using sudo apt update.

shivangini@ubuntu:~$ sudo apt update

Q19. Display the list of currently installed packages on your system using apt list --installed.

shivangini@ubuntu:~$ apt list --installed

Q20. Use chown to change the owner of script.sh to yourself explicitly, and verify with ls -l.

shivangini@ubuntu:~$ sudo chown shivangini script.sh
shivangini@ubuntu:~$ ls -l script.sh

Note: This command assumes your actual Ubuntu username is shivangini.

SECTION B — MEDIUM EXERCISES

Q21. Set permissions on script.sh so that the owner has full access (rwx), the group has read and execute (r-x), and others have zero permissions. Write the numeric command.

shivangini@ubuntu:~$ chmod 750 script.sh
shivangini@ubuntu:~$ ls -l script.sh

Q22. Explain the exact difference in rights between running chmod 700 file and running chmod 007 file.

Command 1:

shivangini@ubuntu:~$ chmod 700 notes.txt
shivangini@ubuntu:~$ ls -l notes.txt

Command 2:

shivangini@ubuntu:~$ chmod 007 notes.txt
shivangini@ubuntu:~$ ls -l notes.txt

Explanation:
chmod 700 gives read, write, and execute permissions to the owner only. Group and others have no permissions.

chmod 007 gives read, write, and execute permissions to others only. The owner and group have no permissions.

Q23. Use chown to change both the user owner and group owner of a file to student:student in a single command.

shivangini@ubuntu:~$ sudo chown student:student notes.txt
shivangini@ubuntu:~$ ls -l notes.txt

Q24. Launch the top interactive process monitor, press P to sort processes by CPU utilization, and press q to exit.

shivangini@ubuntu:~$ top

Press P to sort by CPU usage.
Press q to exit.

Q25. Start two background sleep processes, list them with jobs, and bring job 1 to the foreground using fg.

shivangini@ubuntu:~$ sleep 100 &
shivangini@ubuntu:~$ sleep 200 &
shivangini@ubunt